

BillSG Spec Audit — Change Log

Kiro BuildFest 2026 · Team Amudhan & Raghul
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A deep audit of all 24 Kiro-generated subfolder spec files (`.kiro/specs/*/requirements.md`, `design.md`, `tasks.md`) was conducted against the 5 original committed flat specs and 2 steering docs.

The audit found **10 critical issues** and **10 minor issues**. All 20 were fixed in this session across 12 spec files.

Critical Issues (C1 – C10)

C1 Merdeka Generation year typo + missing second arm

Files: `eligibility-matrix/design.md` `eligibility-matrix/requirements.md`

Before	<code>is_merdeka: boolean // Born 1950–1959, SC before 31 Dec 1986</code>
After	<code>is_merdeka: boolean // Born 1950–1959, SC before 31 Dec 1996 // (or Born ≤ 1949, not Pioneer, SC before 31 Dec 1996)</code>

In requirements.md Req 9, added AC3 documenting both Merdeka arms: (a) Born 1950–1959 AND SC by 31 Dec 1996, OR (b) Born before 1950, not Pioneer, AND SC by 31 Dec 1996.

C2 ElderFund missing CareShield Life / ElderShield exclusion

Files: `eligibility-matrix/design.md` `eligibility-matrix/requirements.md`

Before	<code>Req 13 AC1: "...AND MediSave < \$10,000 AND needs ADL assistance..." design.md item 12: "SC AND age ≥ 30 AND PCHI ≤ 1500"</code>
After	<code>AC1 now includes: "AND NOT a CareShield Life or ElderShield policyholder" design.md item 12 updated to match</code>

C3 ElderFund MediSave balance check missing from design.md & tasks.md

Files: `eligibility-matrix/design.md` `eligibility-matrix/tasks.md`

Before	design.md item 12: "SC AND age $\geq$ 30 AND PCHI $\leq$ 1500" tasks.md Task 7 elderfund.ts: same truncated criteria
After	design.md item 12: added "AND MediSave balance < \$10,000" tasks.md Task 7: bullet updated to include the MediSave balance check

C4 ElderFund ADL qualifier imprecise, payout amount missing

Files: `eligibility-matrix/requirements.md`

Before	AC1: "needs ADL assistance" (no ADL count specified, no payout stated)
After	AC1: "requires full assistance with at least 3 of 6 Activities of Daily Living" New AC3: "GIVEN ElderFund unclaimed THEN payout up to \$250/month (subject to assessment)"

C5 ComCare PR eligibility missing SC family member condition

Files: `eligibility-matrix/design.md` `eligibility-matrix/requirements.md`

Before	design.md item 13: "ComCare – unclaimed if SC/PR AND PCHI $\leq$ 800..." Req 14 AC1: "GIVEN citizenship IN (SC, PR) AND monthly_pchi $\leq$ 800..."
After	design.md item 13: "...for PR: SMTA only, requires at least one SC family member in household" Req 14: split into AC1 (SC rule) and AC2 (PR SMTA with SC family member condition)

C6 MediShield Life "once per policy year" deductible rule missing

Files: `eligibility-matrix/requirements.md`

Before	Req 3 had 6 acceptance criteria; no mention of once-per-year deductible rule
After	Added AC7: "GIVEN MediShield Life deductible WHEN same policy year AND multiple hospital admissions THEN deductible is paid only once per policy year (not per admission)"

C7 bill-explainer ward_class=unknown contradiction between requirements.md and design.md

Files: `bill-explainer/requirements.md`

design.md (Pre-Generation Hook) correctly states: ward_class unknown $\rightarrow$ skip item with fallback message. requirements.md Req 4 AC3 said the opposite — "items still processed". Fixed requirements.md to match design.md.

Before	Req 4 AC3: "GIVEN ward_class = 'unknown' WHEN hook runs THEN items still processed (warning shown separately)"
After	

After

```
Req 4 AC3: "GIVEN ward_class = 'unknown' WHEN hook runs
THEN skip all items with plain_english = 'Could not explain this item.
Please check your original bill.'"
```

C8 scheme-matcher/requirements.md missing all 20 end-to-end scenario test cases

Files: `scheme-matcher/requirements.md`

The original flat spec (`scheme_matcher.md`) contained 20 GIVEN/WHEN/THEN persona scenarios — Uncle Tan (Pioneer), PR age 45, Foreigner, private hospital, PCHI=0, age 15 (MediFund Junior), urgent >\$10K callout, IP rider, Path A, boundary age 18, etc. None were carried over into the new subfolder `requirements.md`.

Before

Req 10 had 9 edge-case ACs; no scenario-based test matrix

After

Added Requirement 11 "Acceptance Test Scenarios" with all 20 GIVEN/WHEN/THEN scenarios restored verbatim from the original flat spec (retrieved via `git show`)

C9 CHAS Orange Annual Value range missing from scraper extracted data

Files: `scraper/design.md`

Before

```
{ "blue_pchi_max": 1500, "blue_av_max": 21000,
  "orange_pchi_min": 1501, "orange_pchi_max": 2300,
  "green_pchi_min": 2301, "applies_to": ["SC"] }
```

After

```
{ "blue_pchi_max": 1500, "blue_av_max": 21000,
  "orange_pchi_min": 1501, "orange_pchi_max": 2300,
  "orange_av_min": 21001, "orange_av_max": 31000,
  "green_pchi_min": 2301, "applies_to": ["SC"] }
```

C10 action-plan dead code in `getActionPlan()` function

Files: `action-plan/design.md`

The second guard `if (scheme_id === 'medifund_junior' && userProfile.age >= 18)` is unreachable: the eligibility matrix only ever assigns `medifund_junior` when `age < 18`. If executed it would silently return the wrong template.

Before

```
if (scheme_id === 'medifund_silver' && userProfile.age < 18) return
  TEMPLATES['medifund_junior']
if (scheme_id === 'medifund_junior' && userProfile.age >= 18) return
  TEMPLATES['medifund'] // dead code
return TEMPLATES[scheme_id] ?? null
```

After

```
if (scheme_id === 'medifund_silver' && userProfile.age < 18) return
  TEMPLATES['medifund_junior']
return TEMPLATES[scheme_id] ?? null
// dead code guard removed – medifund_junior is only assigned when age < 18
```

Minor Issues (M1 – M10)

M1 Government Subsidy effective date not documented

Files: `eligibility-matrix/requirements.md`

Before	Req 2 had no effective date for the subsidy tier structure
After	Added "Effective: 1 October 2024" note to Req 2 user story

M2 ComCare LTA cash amounts not specified

Files: `eligibility-matrix/requirements.md`

Before	Req 14 AC3: "GIVEN LTA type THEN action: Visit any Social Service Office" (no dollar amounts)
After	Added AC4: "GIVEN LTA type THEN cash assistance \$760–\$2,230/month (from April 2025, based on household size and income)"

M3 ComCare above-\$800 PCHI case-by-case note missing from scheme-matcher

Files: `scheme-matcher/requirements.md`

Before	Req 4 AC7: "GIVEN final_payable > 500 AND monthly_pchi <= 1000 THEN flag ComCare" (no mention of case-by-case assessment above \$800)
After	Added AC8: "GIVEN monthly_pchi slightly above \$800 (up to ~\$1,000) THEN still flag ComCare as unclaimed – eligibility is assessed case-by-case; patients near the threshold should still apply"

M4 app/match/page.tsx dual-ownership ambiguity

Files: `scheme-matcher/tasks.md` `ui-flow/tasks.md`

Before	Both Task 9 (scheme-matcher) and Task 4 (ui-flow) listed "Create app/match/page.tsx" with no clarification on which spec owns what
After	Both now include ownership comment: "ui-flow owns the page UI layout; scheme-matcher owns the API route (app/api/match-schemes/route.ts) and all lib/scheme-matcher/ logic"

M5 app/explain/page.tsx dual-ownership ambiguity

Files: `bill-explainer/tasks.md` `ui-flow/tasks.md`

Before	Both Task 5 (bill-explainer) and Task 3 (ui-flow) listed "Create app/explain/page.tsx" with no clarification
After	Both now include ownership comment: "ui-flow owns the page UI layout; bill-explainer owns the API route (app/api/explain-bill/route.ts) and all lib/bill-explainer/ logic"

M6 MediSave secondary source URL missing from scraper

Files: scraper/design.md

Before	No MediSave section in "Extracted Data Points Per Scheme"
After	Added MediSave section with: Primary: moh.gov.sg/managing-expenses/schemes-and-subsidies/medisave/ Secondary: cpf.gov.sg/member/healthcare-financing/medisave (CPF Board is the authoritative source for withdrawal limit tables)

M7 schemes_updated: 11 vs 14 total schemes — ambiguous count

Files: scraper/design.md

Before	Response format: { success: true, schemes_updated: 11, timestamp: ... } (no explanation – could be confused with the 14-scheme total count)
After	Added note: "schemes_updated: 11 refers to the number of government pages scraped (11 URLs), not the total count of healthcare schemes (14)"

M8 Three disclaimer variants look like duplication (they are intentional)

Files: ui-flow/design.md

Before	Three different disclaimer strings listed with no justification for the differences
After	Added note: "The three disclaimer variants are intentionally different – each is sourced from its respective spec and calibrated to that page's context. Do not consolidate into a single disclaimer."

M9 onboarding/tasks.md missing app/api/onboarding/route.ts task

Files: onboarding/tasks.md

Before	Tasks 1–10 covered wizard UI and dashboard only; no server-side session API task
After	Added Task 11: Create app/api/onboarding/route.ts – Accept POST with OnboardingFormState, return mapped UserProfile – Store UserProfile in server-side session (no persistent storage, no PII logged) – Return session ID for subsequent phase API calls

M10 **Scheme count "14" inconsistent with product.md "12+"**

Files: `eligibility-matrix/requirements.md`

Before	Overview: "all 14 Singapore healthcare schemes" (unexplained discrepancy with product.md)
After	Added note: "product.md originally referenced '12+ schemes' – the count expanded to 14 during spec refinement"

Files Modified (12 total)

<code>.kiro/specs/eligibility-matrix/design.md</code>	C1, C2, C3, C4, C5
<code>.kiro/specs/eligibility-matrix/requirements.md</code>	C1, C2, C4, C5, C6, M1, M2, M10
<code>.kiro/specs/eligibility-matrix/tasks.md</code>	C3
<code>.kiro/specs/bill-explainer/requirements.md</code>	C7
<code>.kiro/specs/bill-explainer/tasks.md</code>	M5
<code>.kiro/specs/scheme-matcher/requirements.md</code>	C8, M3
<code>.kiro/specs/scheme-matcher/tasks.md</code>	M4
<code>.kiro/specs/scraper/design.md</code>	C9, M6, M7
<code>.kiro/specs/action-plan/design.md</code>	C10
<code>.kiro/specs/onboarding/tasks.md</code>	M9
<code>.kiro/specs/ui-flow/tasks.md</code>	M4, M5
<code>.kiro/specs/ui-flow/design.md</code>	M8

No changes were made to `.kiro/steering/`, source code, or any file outside `.kiro/specs/`. All Kiro spec linter warnings (missing `## Architecture`, `## Data Models`, etc.) are pre-existing across all spec files and were not introduced by these fixes.